EEE407 Tutorial 2b

1. Lim & Co Law Firm is located at a commercial office building with its business hours is from 8.30 am until 5.30 pm 5 days a week from Monday to Friday. From a desktop energy audit and interview conducted, the electrical equipment that been used together with their power rating and usage time is listed in Table 2-1 below.

a) Based on the information given, calculate the amount of electrical energy used and cost of electricity a month (based on commercial Tariff B: https://www.tnb.com.my/commercial-industrial/pricing-tariffs1)

b) Propose a no cost energy saving approach for their office.

c) Propose a low-cost energy saving approach if some budget is allocated.

Table 2-1

No.	**Electrical Equipment**	**No. of Unit**	**Rated Power**	**Daily Operation Period**
1.	Air Conditioner Unit	4	1.0 HP	9 hours
2.	Air Conditioner Unit	2	2.5 HP	9 hours
3.	Fluorescent Tube Lamp	28	36 W	9 hours
4.	Desktop Computer	10	300 W	9 hours
5.	Printer	5	100 W	2 hours
6.	Photocopier Machine	1	500 W	2 hours
7.	LCD TV	1	150 W	9 hours
8.	Refrigerator	1	500 W	24 hours
9.	Microwave Oven	1	1200 W	1 hour
10.	Water Heater/Cooler Unit	1	700 W	24 hours

2. A scientific lab has replaced its old water-cooled chiller system with a new and more efficient system. The performance of the old chiller system was measured before its removal and it is tabulated in Table 2-2a. Using single linear regression method, plot a graph and find the correlation of the cooling load and electrical power used, and then develop a baseline equation using MS Excel based-on the data gathered.

Table 2-2a

Cooling load (tons)	Electrical power (kW) pre-retrofit
100	80
200	90
300	100
400	150
500	300
600	289
700	322
800	402
900	478
1000	500
1100	688
1200	648
1300	850
1400	893
1500	989

After replacing the old chiller with a new one, the average Cooling Load (Tons) and Electrical Load (kW) was recorded hourly to measure its performance, as tabulated in Table 2-2b, calculate the energy savings.

Table 2-2b

Time	Cooling load (tons)	Electrical load (kW) post-retrofit
8:00	280	80
9:00	380	100
10:00	500	200
11:00	700	220
12:00	880	270
13:00	920	330
14:00	950	380
15:00	890	355
16:00	850	310
17:00	790	280
18:00	400	120
19:00	330	90
20:00	200	50
21:00	230	50

3. Xiamen University has replaced some of the air-conditioning system and lighting systems at its Sepang campus.

a) Energy consumption and cooling degree days (CDD) data were gathered and recorded for 12 months prior to the ECMs implementation. Use single linear regression method to develop a baseline equation form the data gathered during pre-retrofit period as shown in Table 2-3a.

Table 2-3a

Months	CDD	Energy consumption (kWh) pre-retrofit
January	820	1,796,566.00
February	900	1,974,905.00
March	840	1,789,145.00
April	750	1,459,005.00
May	790	1,745,174.00
June	880	1,966,108.00
July	910	1,995,969.00
August	820	1,702,119.00
September	840	1,792,546.00
October	830	1,802,487.00
November	810	1,703,099.00
December	800	1,760,540.00

b) After retrofitting new equipment, energy consumption and cooling degree days (CDD) data were gathered and recorded for another 12 months, as tabulated in Table 2-3b, calculate the cumulative energy savings.

Table 2-3b

Months	Cooling degree days (CDD)	Energy consumption (kWh) post-retrofit
January	830	1,597,642.00
February	910	1,870,300.00
March	850	1,680,655.00
April	790	1,570,778.00
May	800	1,530,622.00
June	890	1,790,497.00
July	950	1,978,738.00
August	830	1,657,045.00
September	850	1,720,307.00
October	840	1,667,156.00
November	780	1,494,829.00
December	790	1,470,013.00